

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course: B.Tech Branch : Mechanical Engineering / Mechanical Engineering(Sandwich)**
/Mechatronics**Semester : VII Subject Code & Name: Name: BTMOE704C Plant Maintenance****Max Marks: 60****Date: 12/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Preventive maintenance consists of the following action. a. Replacement of components b. Repair of components. c. Service of components d. All of the above.	CO1	1
2	Which is known as corrective maintenance? a. Breakdown maintenance b. Scheduled maintenance. c. Predictive maintenance. d. Preventive maintenance	CO1	1
3	Which of the following is an advantage of plant maintenance? a. It decreases plant safety b. It increases life of machinery c. It decreases accuracy of machinery d. It decreases life of machinery	CO4	1
4	What type of maintenance is most effective? a. Shutdown maintenance b. online maintenance c. Corrective maintenance d. Preventing maintenance	CO3	1
5	In reciprocating compressor pressure pressure not increases due to---- a. Suction valves working properly b. Suction line filter choke up c. Suction filter in good condition. d. Delivery valves working properly	CO5	1
6	Shut down maintenance Consist of the following actions. a. Service of component b. Replace if component c. Repair of component d. All of the above	CO4	1
7	----- Maintenance requires monitoring plant equipment health. a. Predictive b. online c. breakdown. d. Corrective	CO4	1
8	What are the various possible causes for a failure of mechanical equipment?	CO5	1

	a. Manufacturing defects	b. Workmanship	c. incorrect usage of equipment	d. All of the above		
9	Due to lake of maintenance, Inventory cost _____				CO4	1
	a. Becomes low	b. Becomes high	c. Remains same	d. All of the above		
10	The Cost Which is done to keep the Plant, Machines & Equipments in proper Working Conditions is known as-----.				CO4	1
	a. Depreciation	b. Maintenance cost	c. Labour cost	d. Material cost		
11	----- techniques is not used in plant condition monitoring				CO6	1
	a. Vibration analysis	b. Casting	c. Ultra sound technology	d. Electrical testing.		
12	-----Is not elements of plant condition monitoring?				CO6	1
	A. Data collection	b. Break down	c. Continuous improvement.	d. Maintenance planning		
Q. 2	Solve the following.					12
A)	Explain briefly different types of plant maintenance?				CO1	6
B)	Explain in detail necessity of shut down maintenance				CO1	6
Q.3	Solve the following.					12
A)	Explain in detail preventive maintenance and its importance.				CO3	6
B)	Explain briefly Online maintenance.				CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Explain repair cycle of plant maintenance.				CO2	6
B)	Explain briefly maintenance of boiler.				CO5	6
C)	Explain Maintenance fault finding and troubleshooting techniques.				CO5	6
Q.5	Solve Any Two of the following.					12
A)	What is importance of stand by unit?				CO3	6
B)	What is mean of trouble shooting?				CO4	6
C)	Write colour code with its specifications				CO2	6
Q. 6	Solve Any Two of the following.					12

A)	Explain different types of technique for condition monitoring	CO6	6
B)	What are the benefits of condition monitoring?	CO4	6
C)	Explain maintenance of pump and compressor briefly.	CO5	6
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